Mispricing Risk Indicator

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Introduction

This project builds a Mispricing Risk Indicator using U.S. financial data from 2015 to 2024. The indicator is inspired by the IMF Working Paper **"Predicting Financial Crises: The Role of Asset Prices" (WP/23/157)** and is designed to monitor periods when asset prices may move away from underlying economic fundamentals.

The model combines six signals across equity markets, market volatility, foreign exchange, fixed income, sovereign spread conditions, and housing prices. Each component is transformed into a comparable percentile-based measure and combined into a **0 to 100 composite risk score**, where higher values indicate greater relative mispricing risk.

Indicator Components

The indicator consists of the following six components:

1. **Real Equity Return** – One-year change in the S&P 500, adjusted for inflation
2. **Equity Market Volatility** – Rolling standard deviation of monthly equity returns
3. **FX Volatility** – Volatility in the USD/EUR exchange rate
4. **Bond Yield Volatility** – Changes in 10-year U.S. Treasury yields
5. **Sovereign FX Spread** – Ratio of emerging market bond dividends to their prices
6. **Real House Price Growth** – One-year change in U.S. house prices, adjusted for inflation

Each period shows a different factor affecting pricing or stability, helping to indicate potential overvaluation.

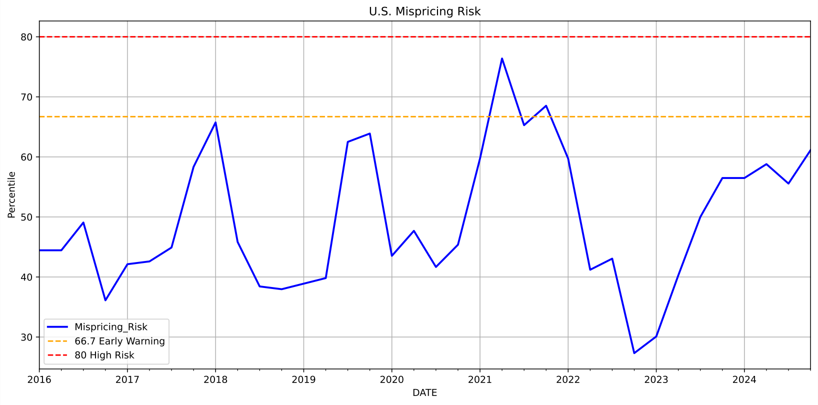
Methodology

The project uses a composite-indicator approach. Each data series is cleaned, converted to monthly frequency, and transformed into a risk-related measure. Since the variables have different units, each component is converted into a percentile score. The final Mispricing Risk Indicator is calculated as the average of the six component scores. The index ranges from **0 to 100**, where higher values indicate higher relative mispricing risk.

Risk zones are defined as follows:

Score Range	Risk Level
0-40	Low
40-66.7	Moderate
66.7-80	Early Warning
80-100	High

Mispricing Risk Plot



Key Drivers by Period

Period	Main Drivers	Reading
2016 to 2019	Equity volatility, bond yield volatility	Stable risk range with brief increases from market uncertainty and rate expectations.
2018 Spike	Equity volatility, bond yield volatility	Risk rose with tighter policy, Treasury yield moves, and trade uncertainty.
2020 to 2021 Peak	Real equity return, housing growth, low rates	Risk increased as equity and housing prices recovered quickly.
2022 Decline	Lower equity momentum, tighter bond conditions	Risk declined as rate hikes tightened financial conditions.
2023 to 2024 Rebound	Equity recovery, housing resilience	Risk began rising again as confidence and asset prices recovered.

Limitations

The Mispricing Risk Indicator is designed as a risk monitoring tool, not as a standalone trading signal or crisis prediction model. A higher score suggests higher relative mispricing risk, but it does not predict the exact timing or direction of market corrections.

The results depend on the selected variables, data frequency, sample period, and proxy measures. Some market risks may not be fully captured, including credit conditions, liquidity stress, investor sentiment, and policy changes. The indicator should therefore be used together with broader macroeconomic and financial analysis.

Conclusion

The project shows how financial time series analysis and percentile scoring can be used to build a practical indicator for monitoring asset pricing risk.

Reference

International Monetary Fund (2023). *Predicting Financial Crises: The Role of Asset Prices*. IMF Working Paper WP/23/157.